IAM FOR IDENTITY MANAEGEMENT

Access Control Report

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1. Objectives / Description:

Students will learn how to:

- Create IAM users and groups
- Assign permissions using policies
- Implement least privilege access
- Test access control using IAM credentials

2. Software's / Tools Required:

- Hardware: PC, Internet access
- Software: AWS Free Tier account (Huzaifa Zahid's account)

3.Lab Environment and Setup:

In this lab we have to setup AWS cloud free tier account on the web.

4.Lab Procedure & Results:

Part 1: Setting up AWS (Free Tier)

After creating the account navigate to the dashboard by typing IAM on the search bar

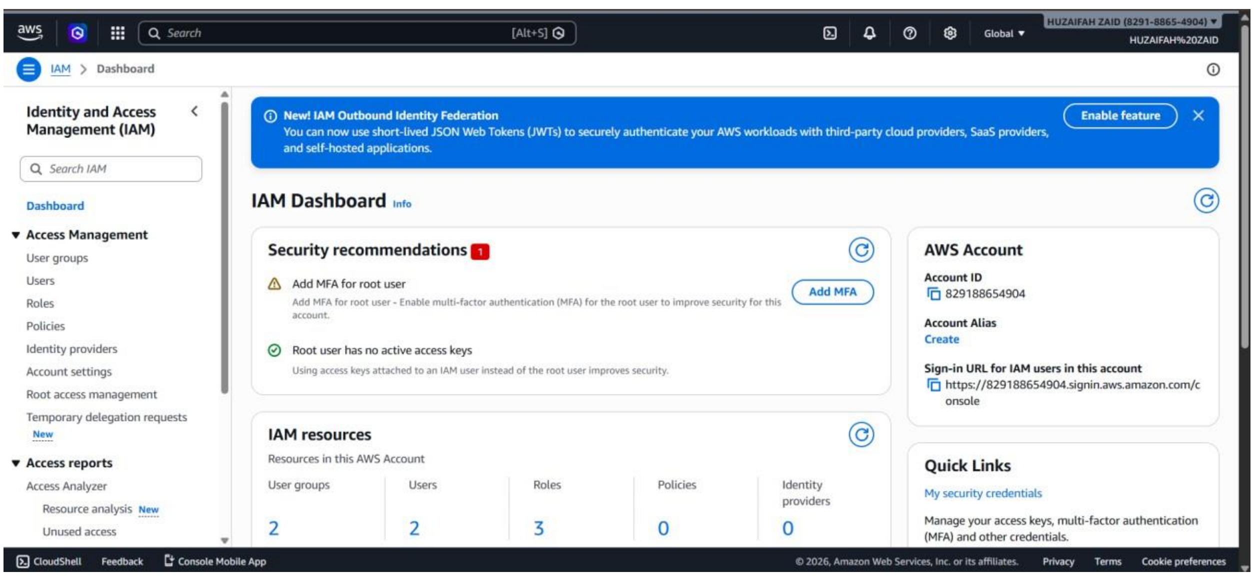


Figure 1

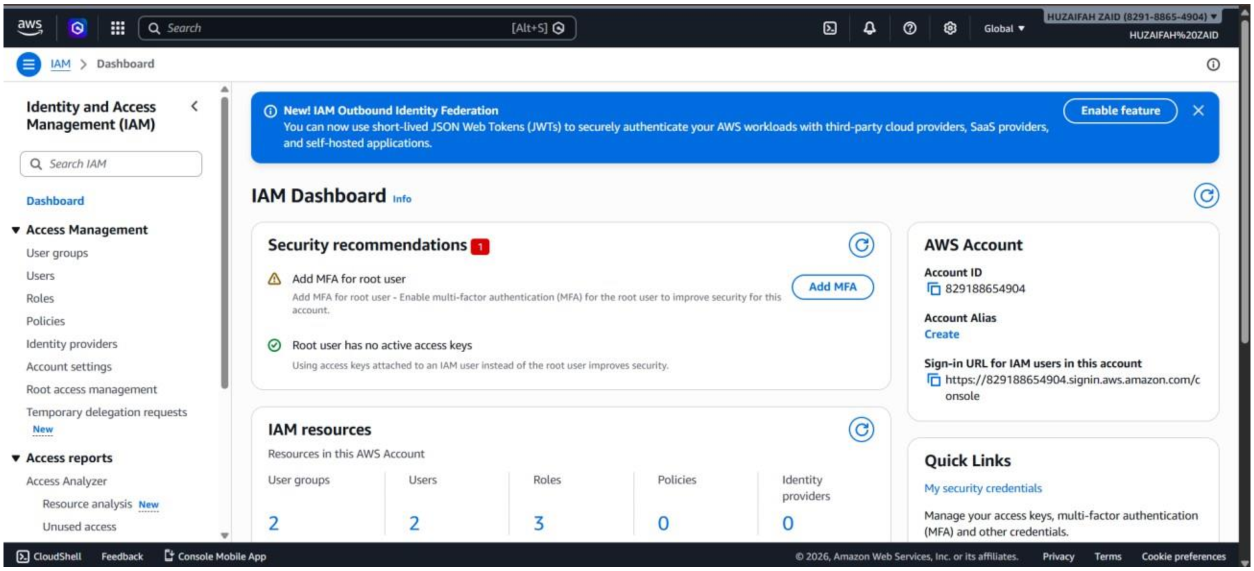
Part 2: Lab Tasks

Task 01 Login to AWS Console:

1. Go to AWS Management Console

2. Login using root account (only for setup)

3. Open IAM Dashboard



Task 02 Create IAM Group:

Navigate to User Groups → Create group

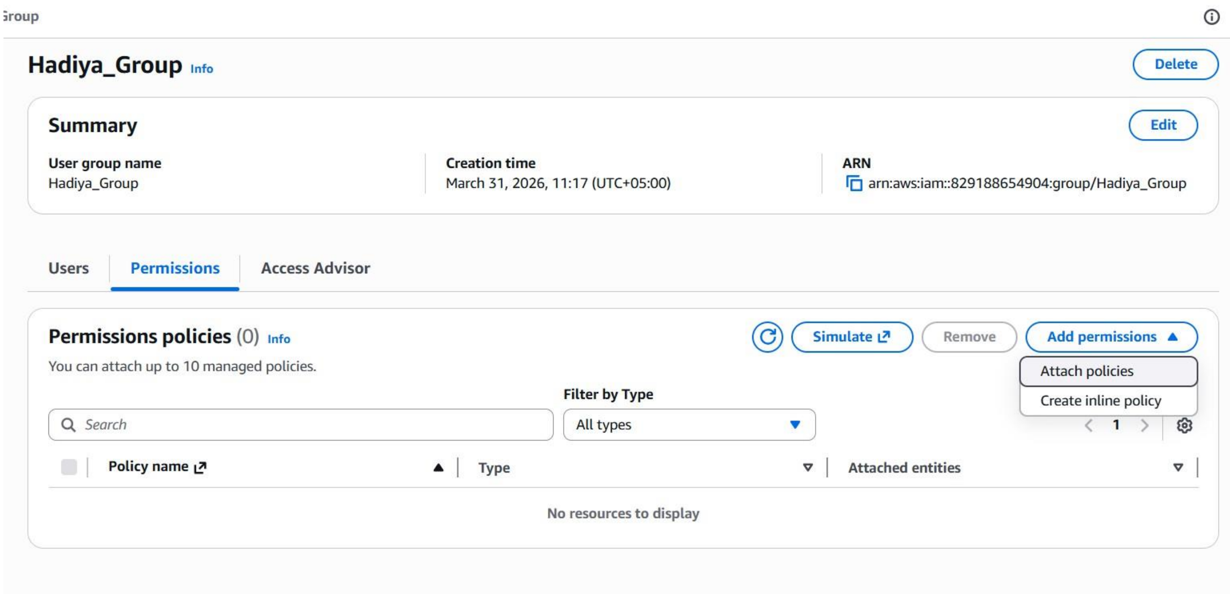
Create two groups:

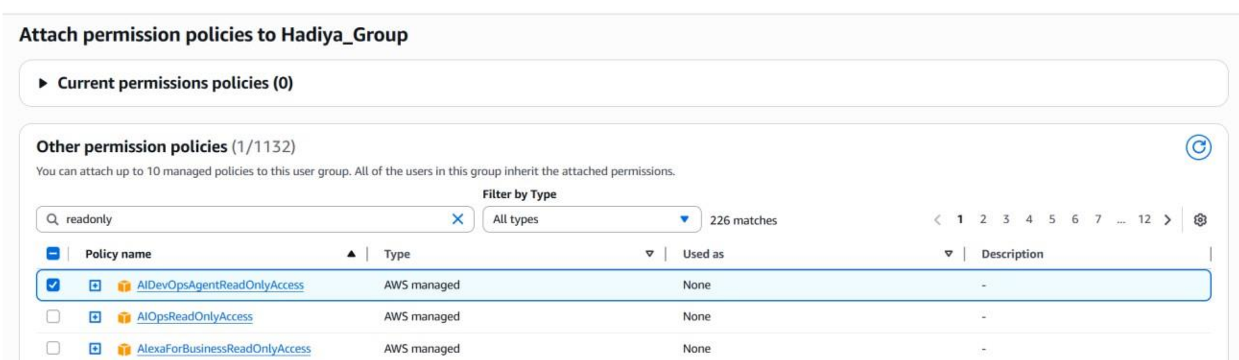
- Students_Group
- Lab_Assistants_Group

Task 03 Attach Policies to Groups:

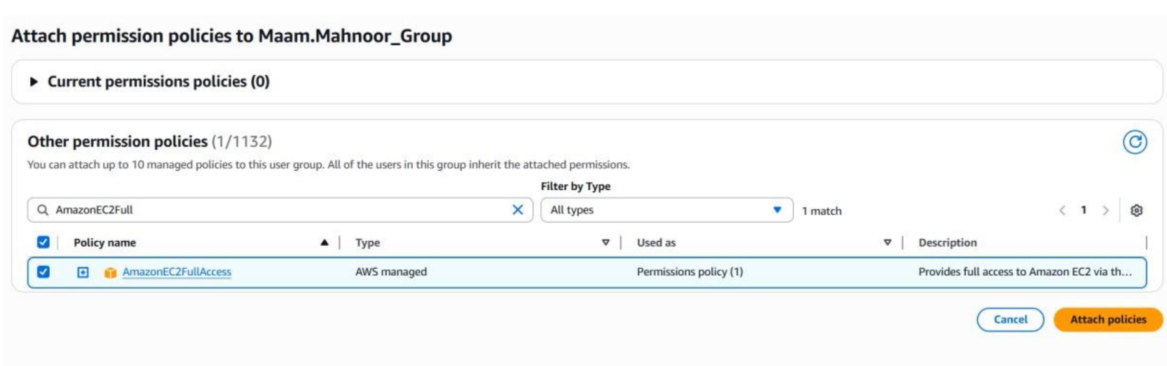
Attach AWS managed policies:

- For Students_Group:
    - ReadOnlyAccess





- For Lab_Assistants_Group:
    - AmazonEC2FullAccess



Task 04 Create IAM Users:

1. Go to Users → Create user

2. Create users:

- student1
- assistant1

3. Enable:

o AWS Management Console access

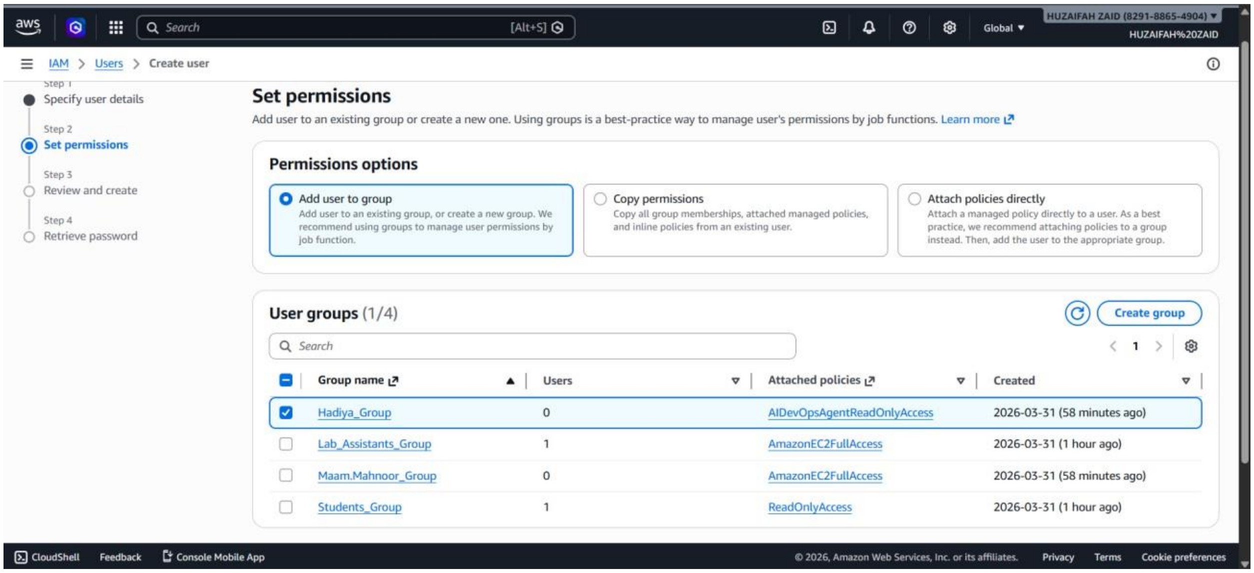
o Set custom password

4. Assign:

o student1 → Students_Group

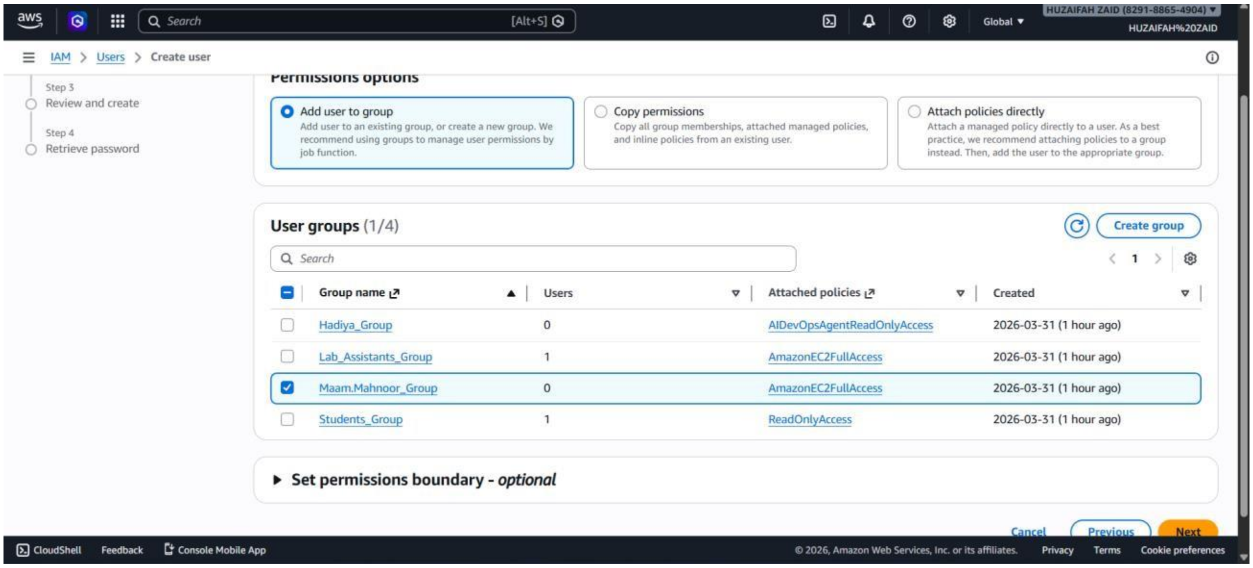
o assistant1 → Lab_Assistants_Group

STUDENT



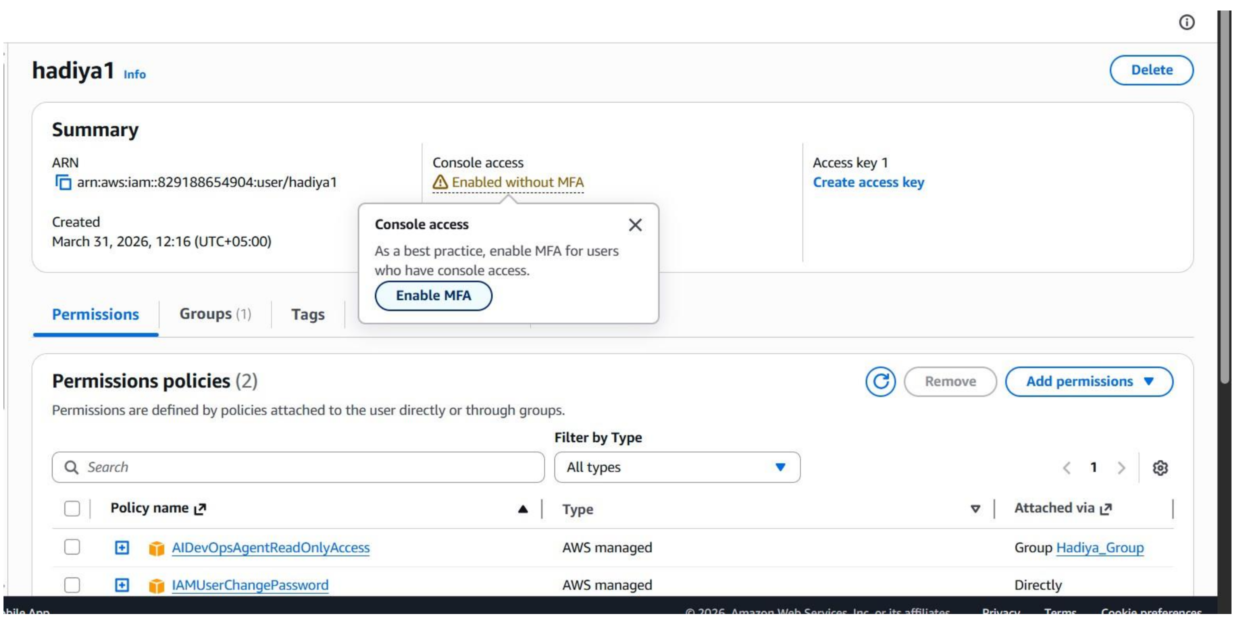
In the same way create for Assiatant

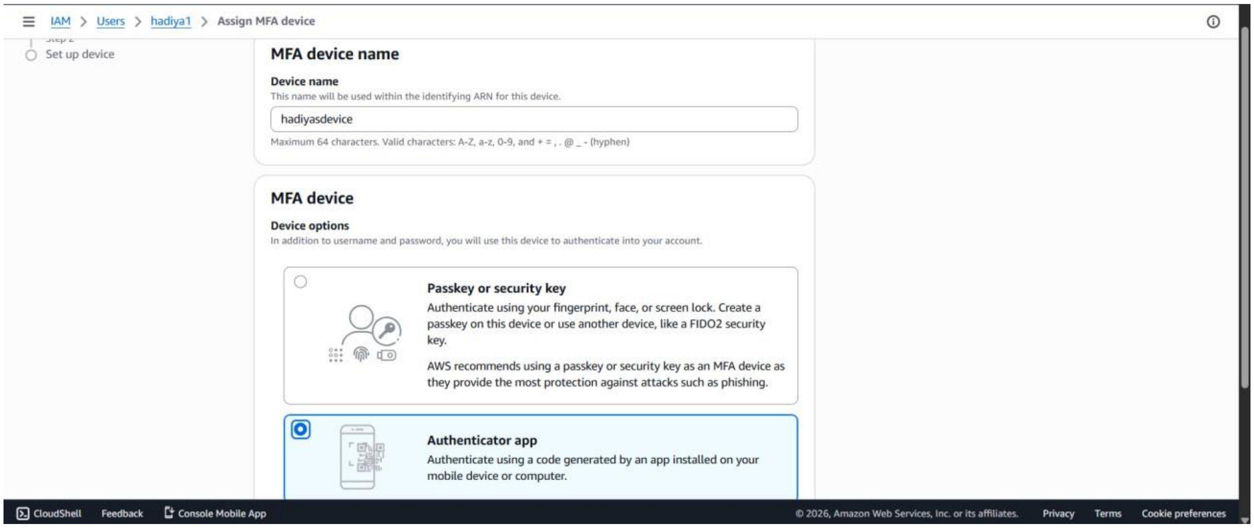
ASSISTANT



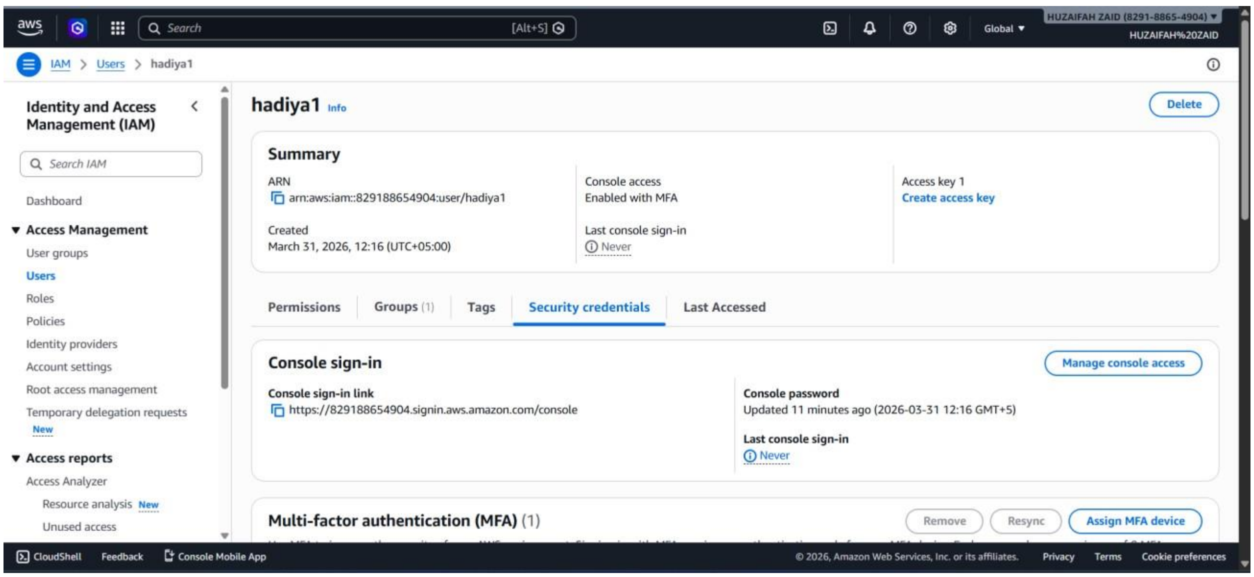
Task 05 Enable MFA (Multi-Factor Authentication)

1. Go to IAM → Users
2. Select a user
3. Enable MFA using:
   - Google Authenticator / Authy

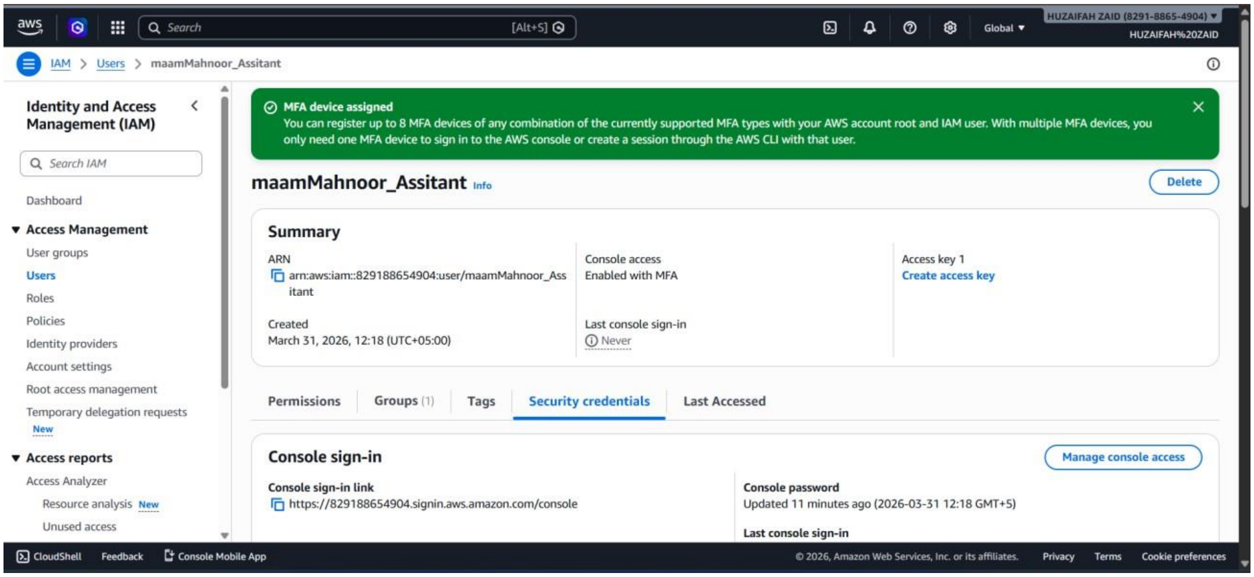




Install Google authenticator on your phone, scan QR code and follow the steps



Do the same for assistant user

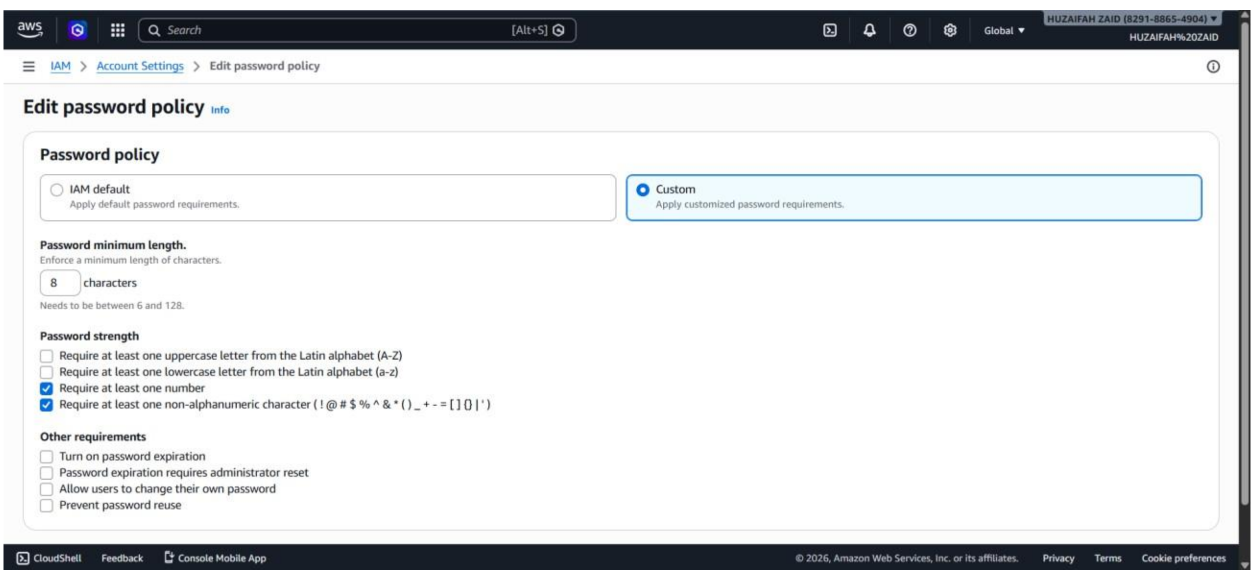


Task 05 Apply Password Policy

1. Go to IAM → Account settings

2. Set:

- Minimum length: 8
- Require numbers & symbols
- Password expiration: optional



Task 8: Explain

Q1. IAM used for?

IAM is used to implement the principle of least privilege—granting only the necessary permissions required for a task. Key uses include:

- User and Group Management: Creating and managing individual users (people or applications) and organizing them into groups to apply shared permissions.
- Role-Based Access Control (RBAC): Creating IAM roles to delegate temporary permissions to trusted entities, such as AWS services (EC2, Lambda), external users, or users from other AWS accounts, eliminating the need for long-term credentials.
- Fine-Grained Permissions: Using JSON policy documents to define granular rules, such as allowing a user to read data from a specific S3 bucket but not delete it.
- Multi-Factor Authentication (MFA): Enforcing an extra layer of protection for users, requiring a password and a one-time code.
- Credential Management: Generating and rotating security credentials (passwords, access keys, SSH keys).
- Access Analysis: Using IAM Access Analyzer to scan for resources shared with external entities and to generate policies based on actual access activity.

Q2. What are other security features provided by AWS?

Beyond IAM, AWS offers a comprehensive suite of security services for infrastructure, data, and application protection:

- **Identity & Access:** AWS IAM Identity Center for centralized access management, and Amazon Cognito for user sign-in in applications.
- **Detection & Monitoring:** Amazon GuardDuty for threat detection, AWS CloudTrail for activity logging, and AWS Security Hub for centralized compliance and security findings. Amazon Inspector and Detective automate vulnerability scanning and investigation.
- **Data Protection & Encryption:** AWS Key Management Service (KMS) for encryption keys, Amazon Macie for protecting sensitive data with ML, and AWS Secrets Manager for secret rotation.
- **Infrastructure & Network Protection:** AWS Shield for DDoS protection, AWS WAF for web application security, and Amazon VPC for network isolation.
- **Governance & Compliance:** AWS Organizations for account management and SCPs, and AWS Artifact for compliance reports.